

Lesson 07.3: Hierarchical Models

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References:

- AIMA (Artificial Intelligence: a Modern Approach)
 - Chap 15: Probabilistic programming
- Martin, Bayesian Analysis with Python, 2018 (2e)



- *Hierarchical Models*

Hierarchical Models

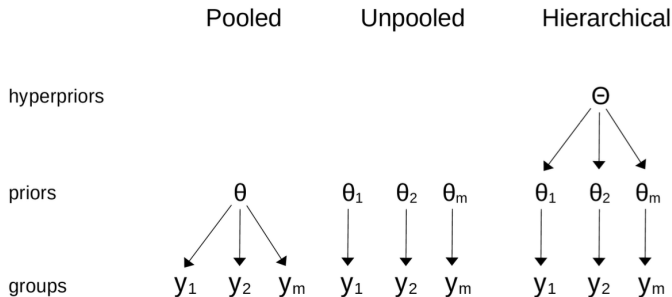
- Aka “multilevel”, “nested”, “mixed-effects” models
- **Key observation:** data points share structure, but also have variations
 - **Group data**
 - * E.g., sales in cities: each city is a market, with common trends
 - **Hierarchical structure**
 - * E.g., students in a school: each student is different, with common factors
 - **Repeated measurements on same objects**
- **Idea:** 💡
 - Model shares information between groups, but allows differences
 - Parameters of prior distributions have a prior distribution
 - * Aka “hyper-priors” (!)
- You can't do this with frequentist approach, only Bayesian approach

Hierarchical Models: Examples

- Many data problems lend themselves to **hierarchical descriptions**
- E.g.,
 - Medical research:
 - * Estimate drug effectiveness
 - * Categorize patients by demographics, disease severity
 - * Estimate cure probability for subgroups
 - Market research
 - * Understand consumer purchasing behavior
 - * Categorize consumers by age, gender, income, education

Unpooled, Pooled, Hierarchical Models

- **Pooled**
 - Groups have the same priors
- **Unpooled**
 - Groups have different priors
- **Hierarchical**
 - Groups have different priors which come from a common prior



Hierarchical Models: Chemical Shift

- **Proteins** are made of 20 amino acids
 - Study proteins with nuclear magnetic resonance
 - Measure “chemical shift”
- **Data** looks like:

	ID	aa	theo	exp	diff
0	1BM8	ILE	61.18	58.27	2.91
1	1BM8	TYR	56.95	56.18	0.77
2	1BM8	SER	56.35	56.84	-0.49
3	1BM8	ALA	51.96	51.01	0.95
4	1BM8	ARG	56.54	54.64	1.90
...	...	...	...	...	...
1771	1KS9	LYS	55.79	57.51	-1.72
1772	1KS9	ARG	58.91	59.02	-0.11
1773	1KS9	LYS	59.49	58.92	0.57
1774	1KS9	GLU	59.48	58.36	1.12
1775	1KS9	SER	58.07	60.55	-2.48

- **ID**: Code of the protein
- **aa**: Name of the amino acid
- **theo**: Theoretical values of chemical shift
- **exp**: Experimental value
- **diff**: Difference between theoretical and experimental value

1776 rows × 5 columns

Hierarchical Models: Chemical Shift

- Given experimental measures of chemical shifts vs theoretical values, evaluate model using different styles
 1. **Pooled**
 - Compute difference between estimates and measures, fit Gaussian
 - More accurate estimates / lose amino acid info
 2. **Unpooled**
 - Fit 20 Gaussians for 20 amino acids
 - Detailed analysis / less accuracy
 3. **Hierarchical**
 - Model groups assuming common population

Chemical Shift: Unpooled Model

- **Model each group independently**
 - Use same model structure to compare groups

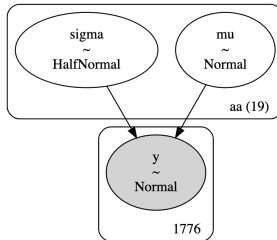
```
In [9]: # Non-hierarchical model.
with pm.Model(coords=coords) as cs_nh:
    # One separate prior for each group.
    mu = pm.Normal('mu', mu=0, sigma=10, dims="aa")
    sigma = pm.HalfNormal("sigma", sigma=10, dims="aa")
    # Likelihood.
    y = pm.Normal("y", mu=mu[idx], sigma=sigma[idx], observed=diff)
    idata_cs_nh = pm.sample()
```

```
Auto-assigning NUTS sampler...
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [mu, sigma]
Output()
```

```
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000) ...
```

```
In [10]: pm.model_to_graphviz(cs_nh)
```

```
Out[10]:
```



Chemical Shift: Hierarchical Model

- **Add two hyperpriors on**

μ

- Mean of μ
- Standard deviation of μ

- **Assume same variance**

σ for all groups

- Modeling choice
- Option to add hyperpriors for σ

- Intermediate situation between single group and 20 separate groups

```
In [12]: with pm.Model(coords=coords) as cs_h:
# Hyper-priors.
mu_mu = pm.Normal("mu_mu", mu=0, sigma=10)
mu_sigma = pm.HalfNormal("mu_sigma", sigma=10)

# Priors.
mu = pm.Normal("mu", mu=mu_mu, sigma=mu_sigma, dims="aa")
sigma = pm.HalfNormal("sigma", sigma=10, dims="aa")

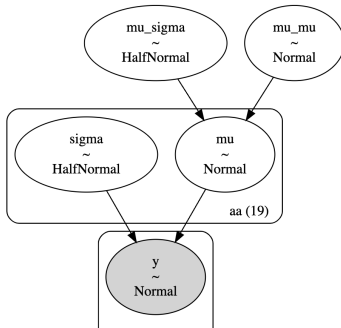
# Likelihood (same as before).
y = pm.Normal("y", mu=mu[idx], sigma=sigma[idx], observed=diff)
idata_cs_h = pm.sample()
```

```
Auto-assigning NUTS sampler...
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [mu_mu, mu_sigma, mu, sigma]
Output()
```

```
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000) ...
```

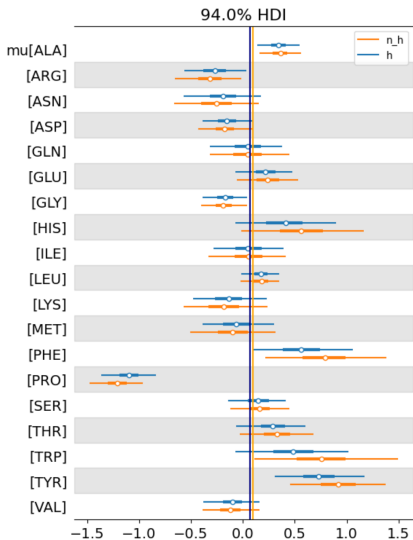
```
In [13]: pm.model_to_graphviz(cs_h)
```

Out[13]:



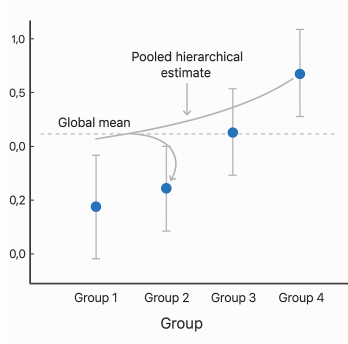
Chemical Shift: Results

- Compare estimates of two models
 - 20 groups
 - Each model has 4 estimated variables
- Plot 94% credible intervals
 - **Blue**: hierarchical
 - **Orange**: non-hierarchical
 - **Black** vertical line: global mean (hierarchical model)
 - **Blue** means pulled towards mean compared to **orange**
- **Shrinkage occurs**



Shrinkage

- **Hierarchical models shrink parameters towards a common mean**
 - Groups share information through the hyper-prior
 - Model groups as neither independent nor a single group
 - Less responsive to extreme values in individual groups
 - Improve estimation for small groups using data from others
- **Amount of shrinkage** depends on data:
 - Groups with more data influence estimates more
 - Similar groups reinforce common estimation
 - Global optimization
- **Result:** inference is more stable



You Need to Know When to Stop

- You can create **hierarchical models with as many levels** as you want
 - **Pros:**
 - * Leverage data structure
 - **Cons**
 - * Don't improve inference quality
 - * Complicate interpretation
- *"Add as many degrees as freedom as needed, but not more than what is warranted" (Occam's razor)*

Tutorial

- [Hierarchical Models](#)